

Theory of Automata (CS3005)

Date: May 20th 2025

Course Instructor(s)

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Bakhtawar, Ms. Abeeها Sattar, Syed Faisal Ali

Final Exam

Total Time (Hrs): 3

Total Marks: 50

Total Questions: 4

Roll No	Section	Student Signature
Do not write below this line		

1. Attempt all the questions.
2. Make clear diagrams with proper notations.
3. Read all the instructions on Answer Sheet, Fill the credentials properly.
4. Do not waste pages. Start from page number 2.
5. Assume and state your assumption clearly if you feel question is having ambiguity.
6. Understanding of the paper is the part of exam, do not call faculty members.

CLO 2: Prove properties of languages and automata with rigorously formal mathematical methods

Question 1: Short Answer questions:

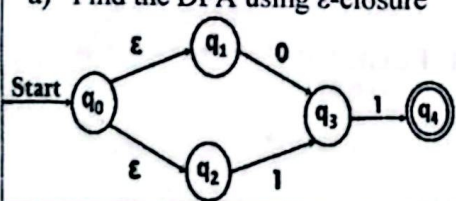
[4 Marks][15 Mins]

- a) Why is the subset construction method used in converting an ϵ -NFA to a DFA?
- b) How does the transition function differ in an NFA compared to a DFA?
- c) Provide a regular expression for Pakistani CNIC.
- d) What is the language of a DFA having no accepting state? What will be the Kleene star of this language?
- e) Differentiate between regular and context-free grammars.
- f) What are the components of a formal grammar?
- g) What is the significance of Kleene's Theorem in automata theory?
- h) Every linear grammar cannot be accepted by DFA, debate in two lines.

CLO 3: Design of automata, RE and CFG

Question 2

[2+1+2+2+5 Marks] [30 mins]

a) Find the DFA using ϵ -closure	b) Give the inverse of your DFA.
	c) Produce the Grammar for the given RE. [[$(0+1)^* 0 (0+1)^* 1 (0+1)^*$] + [$(0+1)^* 1 (0+1)^* 0 (0+1)^*$]]

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CLO 2: Prove properties of languages and automata with rigorously formal mathematical methods

Question 1: Short Answer questions:

[4 Marks] [15 Min]

- Why is the subset construction method used in converting an ϵ -NFA to a DFA?
- How does the transition function differ in an NFA compared to a DFA?
- Provide a regular expression for Pakistani CNIC.
- What is the language of a DFA having no accepting state? What will be the Kleene star of this language?
- Differentiate between regular and context-free grammars.
- What are the components of formal grammar?
- What is the significance of Kleene's Theorem in automata theory?
- Every linear grammar cannot be accepted by DFA, debate in two lines.

Question	Answer
Q1 <u>Why is the subset construction method used in converting an ϵ-NFA to a DFA?</u>	It systematically generates all possible states of the DFA using subsets of NFA states. This ensures the DFA simulates all behaviors of the ϵ -NFA deterministically.
Q2 <u>How does the transition function differ in an NFA compared to a DFA?</u>	In a DFA, the transition function maps a state and input to exactly one next state. In an NFA, it may map to zero, one, or many states. 20.
Q3 <u>Provide a regular expression for Pakistani CNIC.</u>	$[0-9]\{5\} - [0-9]\{7\} - [0-9]\{4\}$ matches the CNIC format like 42201-1234567-1.
Q4 <u>What is the language of a DFA having no accepting state?</u>	The language is empty since no string can reach an accepting state.
Q5 <u>Differentiate between regular and context-free grammars.</u>	Regular grammars generate simpler languages using only linear rules.

Q6
Q7
Q8

Learn points of formal Grammar
Significance of Kleene's Theorem
Spring 2025 School of Computing
Every linear Grammar cannot be accepted by a DFA

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C10.5: Define Turing machines, PDA machines performing simple tasks.

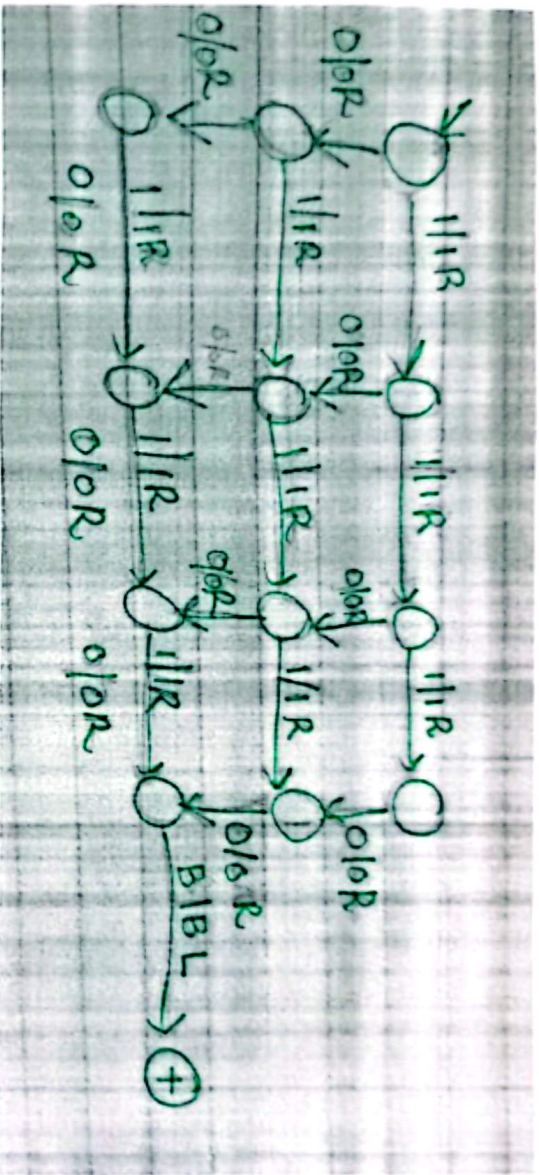
[3+3+5+5 Marks] [60 mins]

Design PDA and CFG of given language.

a) Language of all strings over the alphabet $\{0, 1, 2\}$ that never starts and ends with the same symbol. $S \rightarrow 0A1 0A2 1A0 1A2 2A1 2A0$ $A \rightarrow 0A1 A12A \epsilon$ Note 0, 1, 2 and ϵ will not be strings of the language	b) All strings w such that $w \equiv 3 \pmod 4$ upon the alphabet set $\{a, b\}$. e.g. bbbbaabab, baabbaab,..... b) $S \rightarrow aBbAbBab \mid BaBaBaBa$ $B \rightarrow bB \epsilon$ $A \rightarrow BaBaBaBa \epsilon$ OR $S \rightarrow BaBaBaB$ $B \rightarrow bB \epsilon BaBaBaBaB$
c) $L(A^n) = \{0^m 1^n 0^m \mid n \geq 1, m \geq 1\}$ closure of union of $L_1, L_2, L_3, \dots, L_n$ and $L_1, L_2, L_3, \dots, L_n$ $m \geq 1, n \geq 1$ $L_1 = 0^m 1^n 0^m$ $S = 0S1$ $AB = 1A0$	d) $L = \{0^m 1^n 3^n \leq m \leq 4^n, n \geq 1\}$ $S_1 = 01110111000, 0111000$ $L_1 = \{0^m 1^n m \geq 1, n \geq 1\}$ $L_2 = \{0^m 1^n m \geq 1, n \geq 1\}$ $L_3 = \{0^m 1^n m \geq 1, n \geq 1\}$

a. S-> 0A2/0A1/1A0/1A2/2A0/2A1
A-> 1A/2A/0A/

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b) Accept the language $L = \{b^m a^n a^m b^n \mid m \geq 2n \geq 0\}$
solution

$$n = \frac{1}{2} a^1 b^1, \dots$$

$$b^3 a^1 a^3 b^1$$

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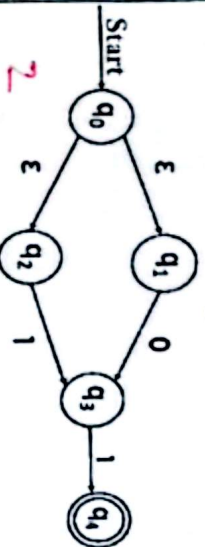
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Q10.3: Design of automata, RE and CFG

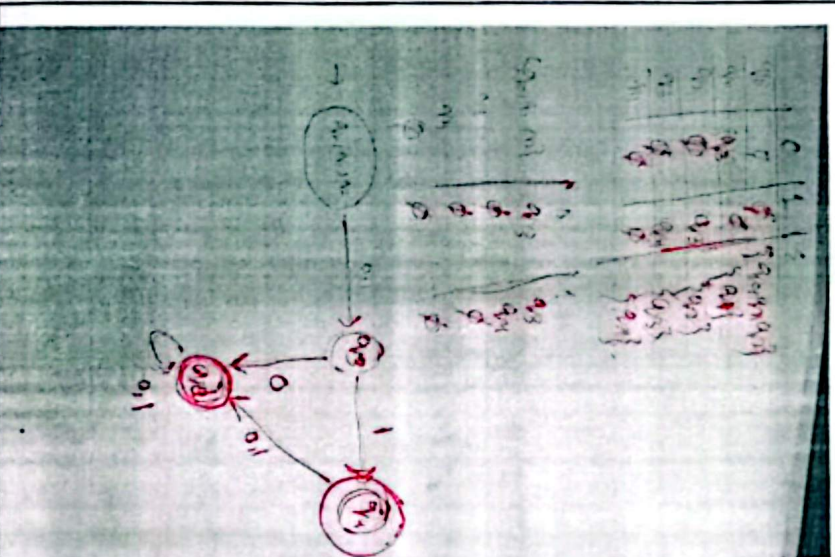
Question 2

[2+1+2+2+5 Marks] | 30 mins

a) Find the DFA using ϵ -closure



solu



b) Give the inverse of your DFA.

solution

Reverse all transitions in the NFA.

inverse is same as complement of machine (final states will be non final, and non final will become final states)

Swap initial and final states:

c) Produce the Grammar for the given RE.

$[(0+1)^*0(0+1)^*1(0+1)^*] + [(0+1)^*1(0+1)^*0(0+1)^*]$

Variables: S, A, B, C

Terminals: 0, 1

Start symbol: S

Productions:

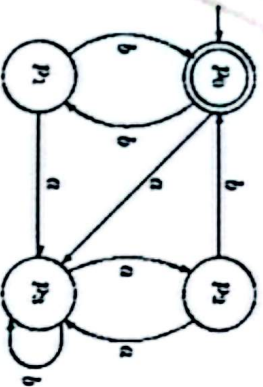
$S \rightarrow A0B1C1A1B0C$

$A \rightarrow 0A1A1C$

$B \rightarrow 0B1B1C$

$C \rightarrow 0C1C1C1C$

d) Produce Grammar of given DFA.



$P_0 \rightarrow aP_3 | bP_1 | \epsilon$
 $P_1 \rightarrow aP_3 | bP_0$
 $P_2 \rightarrow aP_3 | bP_0$
 $P_3 \rightarrow aP_2 | bP_0$

a)

$S \rightarrow 0A | 0A2 | 1A0 | 1A2 | 2A1 | 2A0$

$A \rightarrow 0A | 1A | 2A | \epsilon$

Note 0, 1, 2 and ϵ will not be strings of the language

e) Simplify and clean the following grammar in proper order and then find the CNF:

$S \rightarrow ABCD | AB | CED | FD$

$A \rightarrow abd | bdc | EF$

$B \rightarrow xyz | wxy | xE | yF$

$C \rightarrow xA | yB | wF$

$E \rightarrow a | b | c$

solution

$S \rightarrow ABCD | AB | CED | FD$

$A \rightarrow abd | bdc | EF$

$B \rightarrow xyz | wxy | xE | yF$

$C \rightarrow xA | yB | wF$

$E \rightarrow a | b | c$

① Remove Nonproductions
② Remove unreachable.

$S \rightarrow AB$
 $A \rightarrow abd | bdc$
 $B \rightarrow xyz | wxy | xE$
 $E \rightarrow a | b | c$

No Null or unit productions. ← ③ & ④

CNF

$S \rightarrow AB$

$A \rightarrow ABD | BD | \epsilon$

$B \rightarrow xyz | wxy | xE$

$E \rightarrow a | b | c$

$N_1 \rightarrow BD$

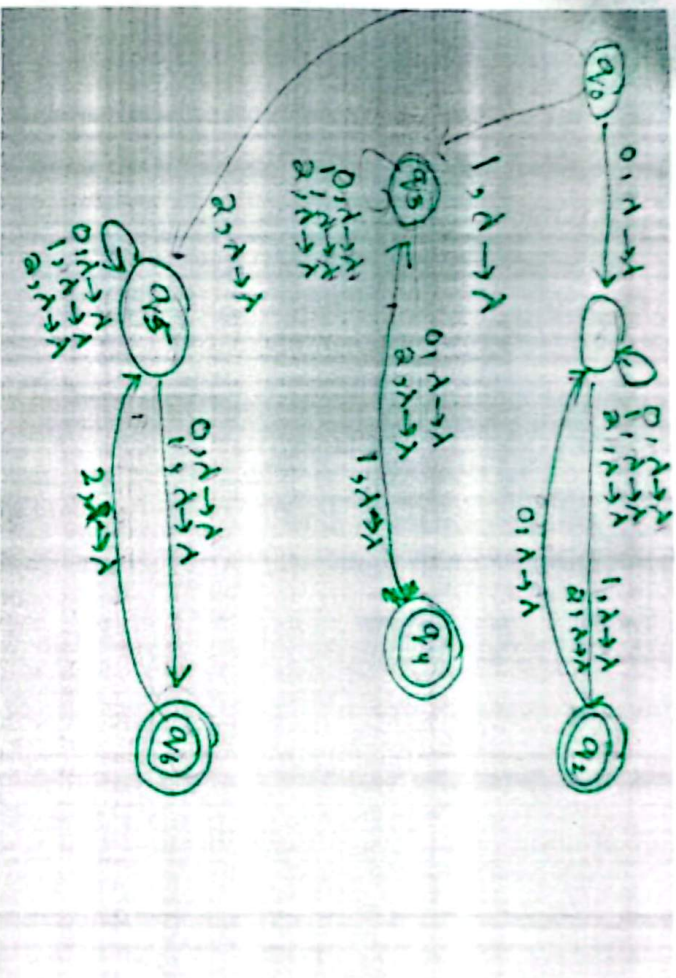
$N_2 \rightarrow x | y$

$A \rightarrow A, N_1, N_2, C$

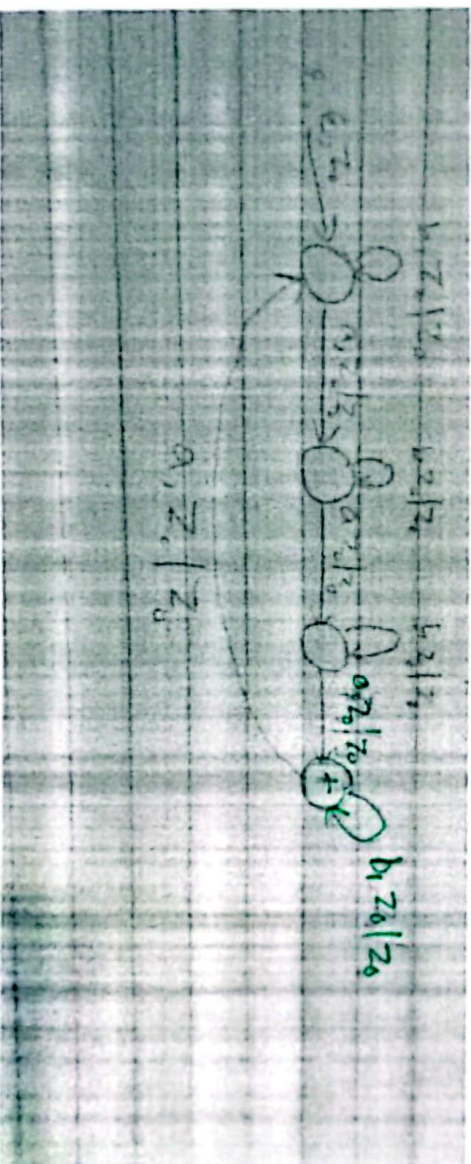
$E \rightarrow N_1 Z_1 | w | N_2 | xE$

CNF.

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b.



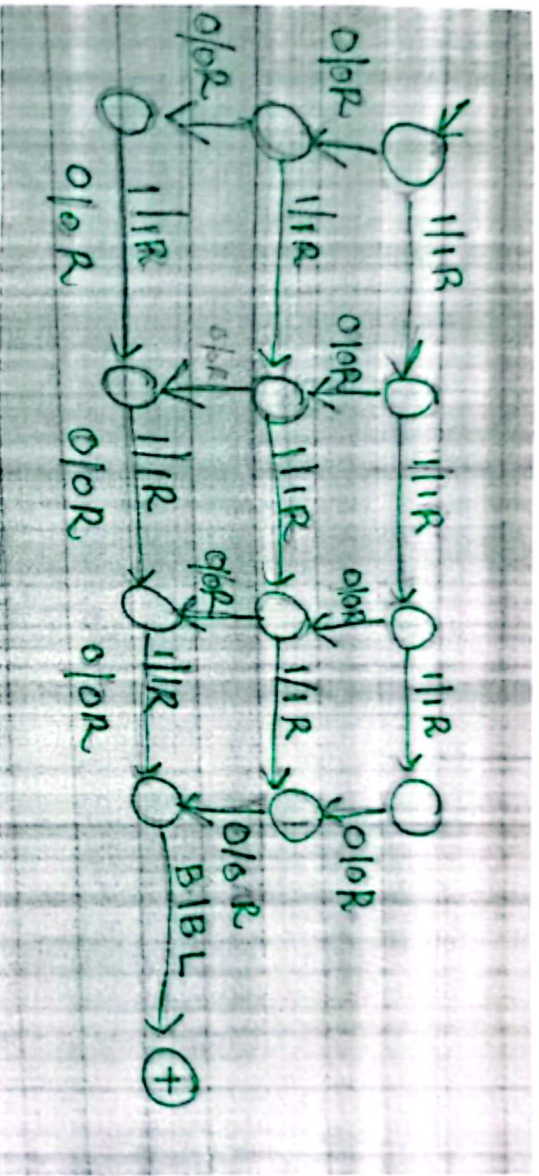
Q10.5: Define Turing machines, PDA machines performing simple tasks.

Question 4.

[4+4+10 Marks] [60 mins]

Design TM for the following descriptions:

- The set of all strings of length 5, defined over $\Sigma = \{0,1\}$ contains at least two 0's or at most three 1's.
Some accepted strings: 00000, 00001, 00111, 10101
Some rejected string: 11110, 11111
solution



b) Accept the language $L = \{b^m a^n a^m b^n \mid m \geq 2n \geq 0\}$ solution

37
11 11
24
-
W
-
5
-
1
1
1

63, 69, 6